Figure 6: Prometheus server configuration

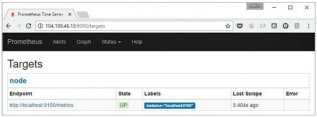

Figure 7: Prometheus server targets configuration

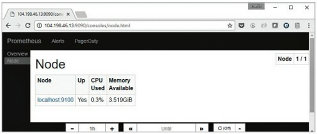

Figure 8: Prometheus server - node list

Figure 9: Node metrics via the visualisation dashboard

which is shown in Figure 5. Just select it in the expressions list and click on the *Execute* button, as shown in the figure.

There is useful information in the *Status* menu option also. You can view your configuration, rules, alerts and targets from there. The default information is shown in Figure 6.

You can click on the *Targets* link in the *Status* main menu to see the targets that you have configured. Since we have configured only one target, that is what we see (Figure 7).

You can view the nodes from the Prometheus UI by visiting */consoles/node.html endpoint* as shown in Figure 8.

You can now click on the *Node* link to see more metrics about the node.

This completes the steps on validating your basic Prometheus set-up. We have only touched upon the surface of this topic. Your next steps should be to assess the several exporters that are available and see which ones address your monitoring requirements, along with any alerts that you would like to set up for your environment.

Cloud Native Computing Foundation (CNCF) and Prometheus

The Cloud Native Computing Foundation (CNCF) is, according to its website, a non-profit organisation committed to advancing the development of cloud native applications and services by creating a new set of common container technologies guided by technical merit and end user value, and inspired by Internet-scale computing.

The first project to be accepted by this foundation was Kubernetes, which is the leading open source solution for container orchestration. Prometheus has been accepted as the second project by this foundation, and that speaks volumes about its functionality and its acceptance as a standard in monitoring applications and microservices.

Prometheus integrates with CNCF's first hosted project, Kubernetes, to support service discovery and monitoring of dynamically scheduled services. Kubernetes also supports Prometheus natively.

Contributing to Prometheus

One of the key reasons for the growth of Prometheus has been the contributions it has received from the community. But often, it is a challenge to understand how you can get started with contributing to some of its core modules. In an article published at NewStack, titled 'Contributing to Prometheus: An Open Source Tutorial', the writer takes up the architecture of the alerts manager and breaks down that module to help us understand how it works and the potential ways in which we can contribute.

Having been accepted by the Cloud Native Computing Foundation, it would help tremendously to contribute to this project, since the visibility of your contribution will be very high. END

References

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